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INTERNATIONAL A-LEVEL PHYSICS

PH03 Fields and their consequences
Report on the examination

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REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS 9630 PH03 JUNE 2023

This paper was accessible to the students and discriminated well, with the mean mark just above 40. The cohort performed, on average, slightly worse in the multiple-choice section (section B) than in section A. As in previous series, the cohort answered calculation-type questions better in general than those requiring an explanation. Also, as in previous series, many students seemed ill-prepared to define quantities. In several questions there was an over-reliance on the equations in the data sheet without a full understanding of what the symbols represented. Question 4 was answered very poorly as it required a higher-level understanding of the link between forces and accelerations.

QUESTION 01

Q01.1 was a derivation question based on a diagram and a given equation. Many students did not appreciate that the command of "*show that*" required them to use the values from the diagram and their knowledge of forces to derive the equation given in the question. Some students acknowledged that the centripetal force was given by $mg \tan \theta$ but did not show the steps they had taken to come to that conclusion; this meant they only had access to one of the three marks. Some students struggled to identify the correct angles and therefore could not resolve the forces as they were resolving for the forces below the slope rather than above.

Q01.2 was a relatively straightforward calculation and determination question. Students were able to calculate the time for car **Q** well. Some students failed to consider that this section of the track was semi-circular and did calculations based on this being a circular track. Alternatively, they did calculations to find speed rather than time which meant they were unable to access the final mark.

Q01.3 This question was a derivation based around moments. Several students failed to understand that R_1 and F_1 were both equal to zero. Those who did appreciate this often skipped the steps required with a "*show that*" command and did not include a moments equation using the values from **Figure 5**. A few students also misunderstood the symbol "w" and tried to substitute in values for ω.

QUESTION 02

Q02.1 was a very straightforward definition question. Generally, students knew the two parts to the definition of simple harmonic motion (SHM). Some students used imprecise language such as "*movement*" or "*motion*" to mean displacement. A noticeable minority used the term "*inversely proportional*" rather than stating that they were proportional but in opposite directions.

Q02.2 was a basic magnetic-fields question and this question was done very well. The most common mistake was students using a right-hand rule and getting their answer reversed.

Q02.3, 02.4 and 02.5 were all straightforward calculations and these were done very well. There were several students who rounded their answers to 02.4 and 02.5 to one significant figure (s.f.) meaning they were unable to gain the final marks for those questions.

Q02.6 required students to sketch a negative cosine curve and to label key points. Most students failed to include values for T and amplitude on the axis of their graphs. Many students did not appreciate that

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this was a $-\cos$ graph. It was still possible for these students to gain half marks when they labelled the key values on each axis, but very few did this.

QUESTION 03

Q03.1 was a four-mark "*show that*" question, so students needed to ensure that they were logically laying out their equations and substitutions. A great many students did not do this and, rather than showing where the equations came from, began from a remembered Kepler's third law equation, then made no substitutions and just wrote down the calculated answer. This meant that they missed marking points. Some students tried to use equations for g, assuming that the value of 9.81 was constant no matter what the radius. Students must remember in "*show that*" command questions they need to include the final answer to at least one more s.f. than is given in the question. This is a quality mark to ensure they have calculated things properly.

Q03.2 was a straightforward calculation but there were many approaches to the question. Students were expected to show their working in full, including the equation into which they were substituting their values. Some students failed to do this so could not access full marks despite getting an answer within range. Several students failed to take into account significant figures leaving their answer as the "*show that*" value of 3100. Their answer should have been to at least one more s.f. than was stated in the question to access the second marking point.

Q03.3 was another calculation question where most students were able to access at least some of the marks. Several did not appreciate that the gravitational potential energy (GPE) was negative and therefore when it was added to the kinetic energy their value for the final answer was incorrect. There was a sizeable minority who tried to calculate GPE using the radius of the Earth rather than the orbital radius of the satellite. A similar number attempted to use GPE = mgh, wrongly assuming that g was constant over such a large range.

Q03.4 was generally answered poorly. Many students did not know all the conditions for a geostationary orbit or showed a real misunderstanding for what this type of orbit entailed. Students struggled to use key vocabulary and were often vague with statements. Using words like "*some*" rather than "*maximum*" did not allow them to access MP1. Several students referred to the satellite being "*in the air*," or having the same orbital radius of the Earth.

QUESTION 04

Q04.1 required students to interpret data from a graph. Overall, students appreciated that they had to find the gradient between points **Q** and **R**. The main error was not appreciating that the line between **Q** and **R** is a curve so the average value could only be taken by reading off points **Q** and **R** and finding a gradient between those points. Some students assumed that the y-coordinate of each was the same value and that one was positive and the other negative; it is not. A small number of students tried to apply *suvat* equations. In question 4, students struggled to understand the difference between mass, weight and resultant force throughout.

In Q04.2 the most common wrong answer was that the weight does not change or that the weight increases as they go upwards. The use of generalised phrases such as "*it varies*" rather than the specific "*it decreases with altitude*" meant that some students could not access the mark as their language was too vague.

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In Q04.3 several students confused the idea of free fall with terminal velocity and described how the weight is equal to the air resistance. Some students also believed that, when in space, Earth's gravitational field strength became zero or that "*there is no gravity in space*."

Q04.4 required students to understand more complex ideas around the sensation of weight and they found this concept very challenging. Several students repeated their answer from 04.2 or 04.3 which could gain them no marks. Some of the most able students identified that the gradient represented the acceleration, and that this was proportional to the additional force experienced by the pilot.

QUESTION 05

Q05.1 and 05.2 required students to read off and interpret data from a graph.
Most students scored both marks on 05.1. A few were able to gain only the first mark due to correctly reading the y-intercept but mis-applying the next step.

In Q05.2 there was more than one way to approach the question and most students were able to achieve at least some marks. Many students did not appreciate the powers of ten on the axis and therefore had a final answer with a power of ten error. Students are still reluctant to include all the data they collect from the graph to calculate a gradient, with very few students making any annotations on the graph at all.

Q05.3 was a straightforward calculation where each step could access a mark even if other steps were incorrect. Most students gained at least one mark on this question.

In Q05.4 the first mark was obtained by many students. Only a very few mentioned the antineutrino, and those who did mostly thought it was absorbed by the body or made a vague statement about its mass/charge. The most common wrong answer was that because beta particles are charged, they will be the only type of particle that can be detected outside of the body. This demonstrated that students did not fully understand the concept of what the question was asking and just stated a fact they knew about beta particles compared to gamma photons.

QUESTION 06

Q06.1 was a calculation which required deeper understanding, and this tripped up students who are heavily reliant on the Formulae and Data booklet. Some students tried to use a radial-field equation and then got stuck and others confused the width of the plate with the distance between the plates. Some students confused E for energy with E for electric field strength.

Q06.2 required an explanation of why capacitance changes when a dielectric is added to a charged plate. Some students assumed that the capacitor was connected to a battery despite there being no mention of this in the question. Many gave vague statements and did not use precise language to explain the concepts.

Q06.3 overall was answered very well. The main mistake students made was to use a capacitor energy equation for the battery leading to an answer of $6500\ \text{J}$. They were mostly able to gain one mark, however, for their calculation of x.

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QUESTION 07

Question 7 was probably a novel context for most students.

Q07.1 was not answered well in general as students were not comfortable with the terminology around electric fields.

Q07.2 was a calculation question, which overall was done very well.

Q07.3 was again done quite well with most students accessing MP1. Rounding too early in the question was an issue for some students, meaning that by the final answer an error had been compounded into their value.

Q07.4 required students to describe the interaction between protons and antiprotons with calculations. Many students were able to access one of the two available marks.

SECTION B

Most of the students answered more than half of the multiple-choice questions correctly. The easiest questions were 8, 10, and 18 where around three-quarters of the cohort got the correct answer.

In all but two of the questions, the most popular response was the correct answer. The questions with significant distractors were 9 and 20. In these questions less than one-third of students selected the right option.

In question 9 the most popular answer was 63 showing that students did not appreciate that the energy of an SHM system oscillates twice in one time period.

Question 12 proved to be the most challenging question on section B with only 6% of students gaining the mark.

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